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Neurocognitive outcomes of ADHD as a comorbidity in pediatric epilepsy: a systematic review

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ABSTRACT

Attention-deficit/hyperactivity disorder (ADHD) is a common neuropsychiatric comorbidity in pediatric epilepsy. Although the cognitive effects of epilepsy are well documented, the impact of ADHD as a comorbidity on neurocognitive outcomes remains unclear. This systematic review evaluates cognitive performance in pediatric patients with both epilepsy and ADHD. Following PRISMA 2020 guidelines, searches were conducted in MEDLINE, EMBASE, Cochrane Library and LILACS. Studies were included if they involved patients aged less than or equal to 18 years diagnosed with both epilepsy and ADHD. Fourteen studies met the inclusion criteria. Methodological quality was assessed using the Newcastle–Ottawa Scale (NOS), ROBINS-I and NIH/NHLBI tools. Results indicated that children with both epilepsy and ADHD exhibited more severe cognitive impairments than those with epilepsy only, ADHD only or healthy controls. These included lower Full-Scale and Verbal IQ scores and more frequent omission and commission errors, as well as increased reaction-time variability in attention tasks. Across studies, domains most affected included perceptual-motor function and executive function, as illustrated in a heat map summarizing the distribution and severity of cognitive impairments. These findings underscore that ADHD comorbidity in pediatric epilepsy is associated with exacerbated neurocognitive deficits, highlighting the need for early identification and tailored interventions to support cognitive development.

1. Introduction

The epilepsies comprise a group of disorders defined by an enduring predisposition of the brain to generate epileptic seizures. Based on the 2017 IAE Classification, childhood-onset epilepsy syndromes are divided into three categories: self-limited focal epilepsies, generalized epilepsies, and developmental and/or epileptic encephalopathies. Epilepsy is the most common pediatric neurological condition, affecting 0.5–1.0 % of children under 16 years of age [1–3].

This disease is associated with numerous neuropsychological comorbidities. Accordingly, cognitive impairment is frequently observed in children with epilepsy and may manifest early in the disease course or even precede the first recognized seizures [3].

Epilepsy often coexists with other neurological conditions, most notably attention-deficit/hyperactivity disorder (ADHD), the most common neurobehavioral disorder in childhood. ADHD affects approximately 8 % of the global pediatric population, and between 22.3 % and 30.7 % of children with epilepsy meet criteria for ADHD. Patients with ADHD exhibit inappropriate levels of inattention, hyperactivity, or impulsivity, which can compromise academic performance, well-being, and social interactions [4–8].

Research indicates that ADHD and epilepsy can have a genetic association, since both are polygenetic disorders, characterized by a combination of common and rare genetic variants playing a role. Moreover, it has been identified that those diseases have associated gene loci and a study suggested a potential genetic basis between ADHD and Genetic Generalized Epilepsy. Notwithstanding, additional studies are needed to explain more about this phenomenon, but these shared genetic and neurobiological factors might explain the overlapping cognitive profiles observed in comorbid cases. [9,10].

The fifth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) classifies ADHD into three presentations: predominantly inattentive, predominantly hyperactive/impulsive, and combined. It also defines six key domains of cognitive function: perceptual-

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motor function, language, executive function, learning and memory, complex attention, and social cognition. Deficits in executive function, a neurocognitive process that supports goal-directed problem solving, may underline many ADHD symptoms, including impaired response inhibition, working-memory deficits, and broader executive-control weaknesses [11,12].

Epilepsy itself is also associated with cognitive impairment, as epileptic activity can contribute to severe intellectual or behavioral disabilities. A history of seizures and repeated episodes of status epilepticus may induce progressive alterations in brain connectivity and cognitive decline over time. Moreover, interictal epileptiform discharges can produce transient cognitive impairment [13].

Despite these findings, the combined impact of epilepsy and ADHD on cognitive outcomes in pediatric patients remains unclear due to a paucity of neuropsychological research. This systematic review aims to evaluate cognitive performance in children with both epilepsy and ADHD, thereby informing management and treatment strategies to enhance patient well-being, given the critical role of cognitive function in quality of life, including learning abilities and social interactions.

2. Methods

2.1. Study design

This systematic review was registered in PROSPERO (ID: CRD420251012631) and conducted according to the PRISMA 2020 guidelines [14]. The research question was structured using the PICO model (Population, Intervention, Comparator or Control, and Outcome) to ensure clarity and objectivity in defining eligibility criteria and in constructing the literature search strategy. Population (P) was defined as pediatric patients (aged less than or equal to 18 years of age) diagnosed with epilepsy; Intervention (I) as the presence of ADHD as a comorbidity; Comparator or Control (C) as pediatric patients with epilepsy without an ADHD diagnosis; and Outcome (O) as cognitive measures (e.g., performance on neuropsychological tests, IQ, executive functions, memory, attention). Studies not meeting these PICO criteria were excluded.

2.2. Search strategy and selection criteria

Four databases (PubMed, Embase, LILACS, and the Cochrane Library) were searched for relevant studies published between January 1, 1985, and January 1, 2025, with the last search conducted on January 22, 2025.

The search strategy combined descriptors such as “child,” “pediatrics,” “attention deficit disorders,” “ADHD,” “epilepsy,” and “cognitive outcomes,” using Boolean operators (AND/OR). The complete search strategies for each database, including search terms, fields, Boolean operators, filters, dates, and the number of duplicates removed, are available on Supplement.

The search strategy was developed with the assistance of a librarian from our institution in collaboration with the review team to ensure transparency and reproducibility. Our inclusion criteria comprised studies including prospective and retrospective clinical studies (observational cohorts, cross-sectional and case-control designs), randomized and non-randomized clinical trials, and systematic reviews with *meta*-analyses. We excluded case reports, conference abstracts, editorials, animal model studies, studies involving participants older than 18 years, other clinical conditions affecting cognitive analysis, studies on epileptic encephalopathies, and studies that did not assess relevant cognitive outcomes.

2.3. Study selection

Retrieved records were imported into the Rayyan platform (https://www.rayyan.ai) for screening and duplication. Duplicate studies were removed, and two reviewers (L.Z.L. and I.K.) independently screened titles and abstracts, followed by full-text review. Selection was guided by the PICO framework and the predefined eligibility criteria. The discrepancies were solved by consensus between the reviewers (L.Z.L. and I.K.) followed by consultation with a third reviewer (N.D.P).

A stage-by-stage breakdown of the screening and exclusion process, as well as a list of excluded full-text studies with reasons for exclusion, is available in Supplement.

2.4. Data extraction

Data extraction from the selected studies was conducted using a spreadsheet on the Google Sheets platform. All extracted data were independently reviewed and verified by consensus between two reviewers (L.Z.L., I.K., C.H.S., Y.A.S.). The extracted information included study details (location, study design), number of participants, type of epilepsy, ADHD diagnostic criteria, ADHD subtypes, methods used for cognitive assessment, results, and conclusions. Only data from the selected studies published in the databases were included.

Because pharmacological treatment, particularly with psychostimulants such as methylphenidate, may significantly influence neuropsychological outcomes, information regarding medication use was systematically extracted from each study.

2.5. Quality assessment

The methodological quality of the included studies was assessed using standardized tools according to study design. Cohort and cross-sectional studies were evaluated with the Newcastle–Ottawa Scale (NOS). Additionally, cohort and case–control studies were assessed using the ROBINS-I tool for non-randomized studies, whereas cross-sectional studies were evaluated using the NIH/NHLBI Quality Assessment Tool. Inter-rater reliability for the ROBINS-I and NIH/NHLBI assessments was calculated using Cohen’s kappa coefficient (κ), indicating moderate agreement between reviewers ($\kappa = 0.50$ for ROBINS-I; $\kappa = 0.44$ for NIH/NHLBI). To resolve discrepancies and reach a final decision, a senior author reviewed the items in question (M.L.N, T.W.V).

2.6. Outcome measures

The primary outcomes evaluated in this review were cognitive and executive functions in pediatric patients with epilepsy and comorbid ADHD. Due to the wide range of assessment tools employed across the included studies, no restrictions were imposed on the instruments used. The evaluated domains encompassed cognition, attention, executive functioning, memory, language, behavior, development, emotional regulation, and academic performance. Given the heterogeneity of assessment methods and study designs, a qualitative synthesis of the findings was conducted.

2.7. Procedures for data synthesis

To integrate cognitive outcomes across the included studies, a heat map was developed to visually represent the neuropsychological performance profiles of the different clinical groups. Neuropsychological instruments that directly assessed cognitive performance were first identified, including the Wechsler Intelligence Scale for Children (WISC; WISC-R, WISC-III, WISC-IV), the Wechsler Adult Intelligence Scale–Fourth Edition (WAIS-IV), and the Wechsler Abbreviated Scale of Intelligence (WASI). These instruments were used in 11 of the 14 studies included in the present review.

The six cognitive domains defined by the DSM-5 (complex attention, executive function, learning and memory, language, perceptual–motor function, and social cognition) were mapped onto the corresponding subtests and indices of each instrument. This correspondence was established through structural analysis of each test and conceptual

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interpretation based on DSM-5 definitions. [11].

Quantitative data (mean ± SD) were extracted for the clinical groups of interest: epilepsy with comorbid ADHD, epilepsy only, ADHD only, and healthy controls, when available. The study by Socanski et al. (2013) [15] was excluded from this step because it did not report quantitative test results. To ensure comparability across studies using different versions of Wechsler instruments, all results were converted into qualitative categories of cognitive performance according to the normative criteria described in the WISC-V Interpretive Report [16]. Subtest scores, expressed as scaled scores (mean = 10, SD = 3), and composite indices, expressed as standard scores (mean = 100, SD = 15), were converted into qualitative levels of cognitive functioning. Extremely low scores (≤ 4 or ≤ 69, respectively) were interpreted as indicating severe impairment; scores between 5 and 6 or 70–79 were classified as moderate impairment; scores between 7 and 8 or 80–89 corresponded to mild impairment; and scores equal to or greater than (9 or 90) were considered preserved. The qualitative synthesis design was chosen over *meta*-analysis due to heterogeneity in assessment tools and outcome reporting.

3. Results

3.1. Selection

Out of the 860 articles found through database searches and exclusion of duplicates, 739 articles were analyzed. The PRISMA 2020 flow diagram is presented in Fig. 1 and The PRISMA 2020 Checklist is available in the supplementary materials. Ultimately, 14 articles were included in this review. Quality assessment was not an exclusion criterion but rather a variable that was evaluated.

3.2. Study characteristics

The publication years ranged from 1999 to 2024. No articles published in 2025 met the inclusion criteria at the time of the search. The studies originated from the United States [17–20], South Korea [21–23], Brazil [24,25], Turkey [26,27], Taiwan [28], and Norway [15]. Additionally, one study was a collaborative effort between Switzerland and Germany [29]. All included articles were cross-sectional, case-control, or cohort studies.

A total of 2,332 participants aged between 8 and 18 years were analyzed, of whom 405 had a diagnosis of both epilepsy and ADHD. Regarding epilepsy subtypes, six studies did not specify the type of epilepsy, four included only one specific subtype, and five included more

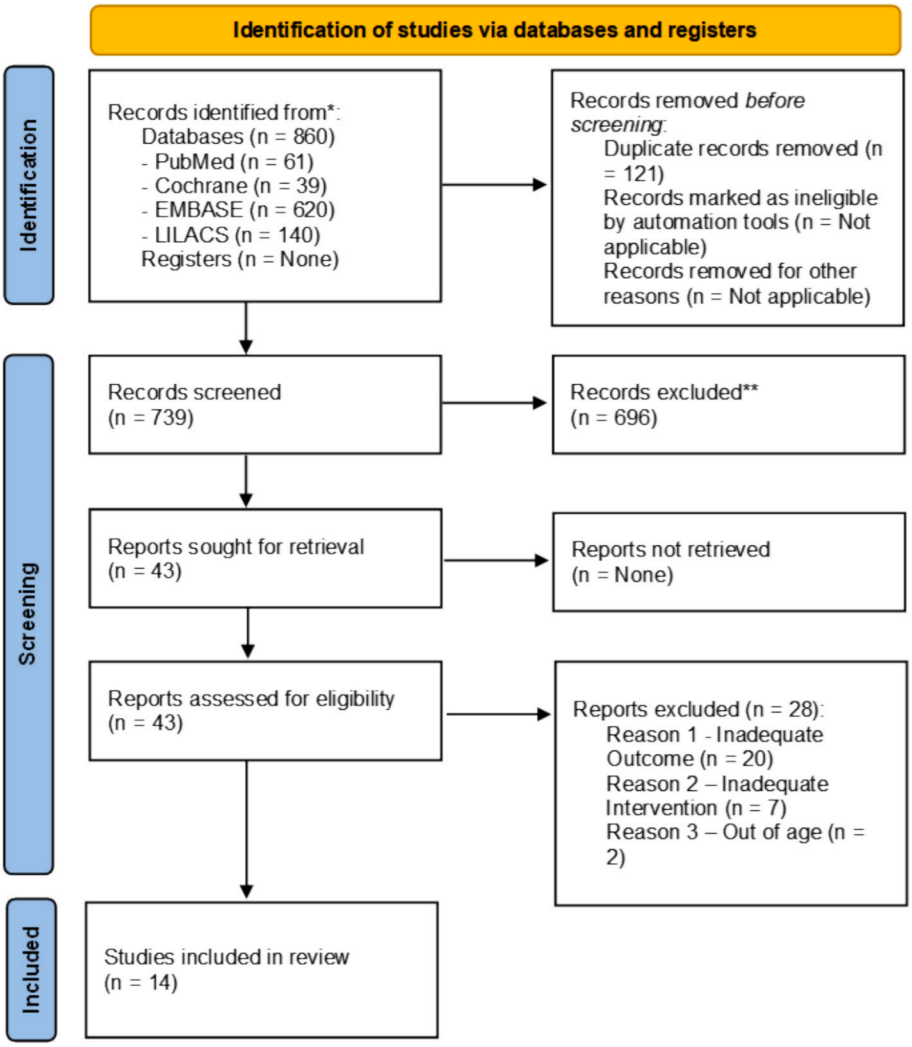


Fig. 1. PRISMA Flow Diagram of Study Selection. Legend: Flow diagram detailing the study selection process following the PRISMA 2020 guidelines. The diagram illustrates the number of records identified, screened, assessed for eligibility, and included in the final review, with reasons for exclusions.

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than one subtype. Notably, the study by Sung-Han Kang (2015) [23] was the only one to perform subgroup analyses based on epilepsy type, distinguishing between generalized and focal seizures.

Most studies included all ADHD subtypes without conducting subgroup analyses. Five studies focused on a single subtype, either the inattentive or the combined type. The studies by Çam Ray (2019) [27] and MacAllister (2012) [19] conducted ADHD subtype analyses specifically in the results of the Stroop TBAG form. Regarding diagnostic criteria, 12 studies employed DSM, with only one using the third edition (DSM-III) instead of the fourth (DSM-IV). Among the remaining studies, one used the DSM-IV in combination with the SNAP-IV, another used the ADHD Rating Scale along with the ADS, and one relied on a pre-existing clinical diagnosis established prior to the study.

The studies used various methods to assess cognitive function. Executive functions were most commonly evaluated using the Wechsler Intelligence Scale for Children (WISC) or Raven's Progressive Matrices (RPM). In contrast, attention was assessed using more heterogeneous instruments, with the most frequently used being the Continuous Performance Test (CPT), the Conners Rating Scale, and the Stroop TBAG form. Table 1 presents the characteristics of the included studies, such as identification, sample, country, Risk of Bias score (RoB), study design, age range, ADHD and Epilepsy subtypes, cognitive evaluation tools used and the main results.

A summary of the main cognitive findings for the group of interest, children and adolescents with epilepsy and ADHD, is presented in Table 2. This table summarizes results from the nine studies that assessed DSM-5 cognitive domains using Wechsler-based instruments (WISC, WAIS-IV, or WASI). The certainty of the evidence for each outcome was evaluated using the GRADE approach. This Summary of Findings provides a concise overview of the Wechsler-based results and serves as a framework for the subsequent domain-specific analyses.

The types of medications administered, the reported treatment status, whether participants were under active medication at the time of cognitive testing, and whether the studies controlled for these effects in their analyses are summarized in Table 3. This information was used to identify and interpret potential confounding influences on cognitive performance outcomes.

The risk of bias assessment for the included non-randomized studies, as evaluated with the ROBINS-I tool is provided on Fig. 2.

The cognitive domains defined by the DSM-5 and the corresponding subtests and indices of each instrument were provided in Table 4.

3.3. Association between cognitive outcomes in patients with epilepsy and ADHD

Executive function evaluation

Although Costa (2015) [24] did not find significant results using Raven's Progressive Matrices (RPM), in the study by Bechtel (2012) [29], the healthy control group performed significantly better in the 2-back task ($p < 0.05$) and the 3-back task ($p = 0.057$) compared to the epilepsy and ADHD group.

Lee (2018) [22], Hermann (2007) [20], Hermann (2008) [18], and MacAllister (2012) [19] reported significant findings using the Wechsler Intelligence Scale for Children (WISC), indicating poorer performance among participants with both epilepsy and ADHD compared to other groups. In the Lee (2018) [22] study, patients with epilepsy and ADHD had lower Full-Scale IQ and Verbal IQ scores compared to those with epilepsy only ($p = 0.014$ and $p = 0.011$, respectively). Similarly, MacAllister (2012) [19] found lower performance in IQ, verbal comprehension, intelligence, and reasoning scores in the epilepsy and ADHD group, regardless of ADHD subtype. Hermann (2007) [20] observed significantly lower scores in executive function, motor function, and parent-reported executive function in the epilepsy and ADHD group compared to the epilepsy-only and healthy control groups (all p-values < 0.001). Likewise, in Hermann (2008) [18], similar results were reported, with significantly worse performance across all cognitive domains in the neurobehavioral comorbidity (ADHD and/or academic problems) group compared to the epilepsy-only and healthy control groups (all p-values < 0.0025), even after adjustment. In contrast, Socanski (2013) [15] did not find any significant differences in executive function based on WISC scores.

Additionally, MacAllister (2012) [19] reported poorer performance in the epilepsy and ADHD groups on the Digit Span Forward Test compared to those with ADHD only. However, in the Digit Span Backward Test, subgroup analysis by ADHD subtype revealed no differences between ADHD combined-type groups ($p = 0.09$).

Attention evaluation

A variety of scales were used to evaluate the attention domain. The Conners' Continuous Performance Test (CPT) revealed poorer performance among the epilepsy and ADHD group in the studies by Liu (2011) [28], Hermann (2008) [18], and Lima (2018) [25]. Liu (2011) [28] reported increased variation in reaction time, including slower Hit Reaction Time, higher Hit Reaction Time Standard Error, and impaired vigilance based on higher scores in the Hit Reaction Time Inter-Stimulus Interval Change, compared to the epilepsy-only and healthy control groups. Similarly, Lima (2018) [25] found worse performance in CPT reaction time standard error ($p = 0.008$), CPT variability ($p = 0.033$), number of perseverations ($p = 0.044$), and interstimulus reaction time ($p = 0.016$) in the epilepsy and ADHD group. Differently, Hermann (2008) [18] analyzed group-by-time interaction for CPT omission (inattention) and commission (impulsivity) errors. While the healthy control and epilepsy-only groups showed decreased errors over time, reflecting improved performance, the epilepsy and comorbidity group showed a slight increase in errors ($p = 0.03$).

Another frequently used scale was the Stroop TBAG form. In Çam Ray (2019) [27], the mean Stroop total error + correction (SEC) SEC-1 and SEC-3 scores in the ADHD and epilepsy group were significantly higher than those of the ADHD-only group (SEC-1: $p = 0.012$; SEC-3: $p = 0.004$). Among ADHD subtypes within the epilepsy group, the mean ST-3 and SEC-3 scores for the inattentive subtype were significantly higher than those for the combined subtype ($p = 0.037$ and $p = 0.036$, respectively). Moreover, mean SEC-1, SEC-2, and SEC-3 scores were higher in the epilepsy and ADHD-combined group compared to the ADHD-combined-only group ($p = 0.016$; $p = 0.041$; $p = 0.024$). Orak (2024) [26] reported similar findings, with significantly longer response times in Stroop 1 ($p < 0.001$), Stroop 2 ($p < 0.001$), Stroop 4 ($p = 0.042$), and total time ($p = 0.012$) in the epilepsy and ADHD group, suggesting impairments in sustained attention, inhibitory control, and impulsivity.

Park (2021) [21] found significantly higher auditory commission error scores in the epilepsy and ADHD group on the Advanced Test of Attention (ATA) compared to the epilepsy-only group ($p = 0.026$). Significant differences in reaction time and reaction time variability were also observed between both groups and healthy controls ($p = 0.008$ and $p = 0.028$). In Lee (2018) [22], the Visual and Auditory ADHD Diagnostic System (ADS) also indicated higher auditory omission error scores in the epilepsy and ADHD group ($p = 0.017$).

Sung-Han Kang (2015) [23] used the Comprehensive Attention Test (CAT) and found lower omission error scores on sustained attention ($p = 0.050$) and flanker tasks ($p = 0.003$), as well as lower forward and backward working memory scores in the spatial working memory task ($p = 0.025$ and $p = 0.035$, respectively) in patients with generalized seizures and ADHD compared to non-ADHD counterparts. In self limited Childhood Epilepsy with Centrotemporal Spikes (SeLECTs) patients, those with ADHD had higher auditory selection task commission error scores ($p = 0.004$) and higher omission error scores on sustained attention tasks ($p = 0.039$) compared to SeLECTs −only patients.

In the study by Semrud-Clikeman (1999) [17], which included only patients with focal impaired awareness seizures (FIAS), the Test of Variables of Attention (TOVA) showed significantly poorer performance in all TOVA indices among the epilepsy and ADHD group ($p = 0.01$). Compared to the ADHD-only group, this group continued to perform

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Table 1
Characteristics of the included studies in the systematic review.

Results	Healthy controls outperformed unmedicated epilepsy/ADHD patients on the 2-back task ($p < 0.05$), with a trend on the 3-back ($p = 0.057$). Developmental ADHD patients had significant deficits on both tasks ($p < 0.01$ and $p < 0.05$). Medication improved 3-back task performance ($p < 0.05$).	Children with epilepsy showed more omission errors ($p = 0.020$), slower reaction times ($p = 0.032$), and higher signal detection scores ($p = 0.007$). The epilepsy + ADHD group had greater RT variability ($p < 0.05$) and reduced vigilance.	No significant differences between epilepsy patients with and without ADHD, except lower WISC Coding and Symbol Search scores in the ADHD group.	The epilepsy + ADHD group had higher auditory commission errors ($p < 0.05$) and elevated SRS scores across all subscales except Social Awareness (all $p < 0.001$).	Patients with ADHD had lower full-scale ($p = 0.014$) and verbal IQ ($p = 0.011$), more auditory omission errors ($p = 0.017$), and higher depressive symptoms ($p = 0.031$).	ADHD patients showed poorer auditory selective attention ($p < 0.020$), flanker task ($p = 0.031$), and working memory (forward $p = 0.007$; backward $p = 0.006$). In generalized seizures + ADHD, omission errors were higher ($p = 0.050$; $p = 0.003$), and working memory lower (forward $p = 0.025$; backward $p = 0.035$). SeLECTs + ADHD showed more commission ($p = 0.004$) and omission errors ($p = 0.039$).	CPS/ADHD patients performed worse than all groups on TOVA indices ($p = 0.01$); CPS and ADHD groups performed worse than controls except for commission errors ($p = 0.03$).
Cognitive Evaluation Tool	RPM | N-back task | Conners Rating Scale	WISC-III; Conneŕs CPT II.	WISC-III; RPM; SAT; TAVIS-III; SPT.	ADHD Rating Scale (ARS) | Advanced Test of Attention (ATA) | CBCL | Children's Sleep Habits Questionnaire (CSHQ) | Disruptive Behavior Disorder Scale (DBD) | Social Responsiveness Scale (SRS) |	WISC | SCWT | Children's Color Trails Test | Children's Depression Inventory | STAI-C	WISC; CAT.	WISC-R; Verbal, performance and full-scale IQs (FSIQs) | Conners Rating Scale | Test of Variables of Attention (TOVA)
Epilepsy Types	Not available	Generalized tonic-clonic seizure | FIAS | Simple partial seizure	Not available	Not available	Juvenile myoclonic epilepsy (8) | Childhood absence epilepsy (6) | Juvenile absence epilepsy (1) | Other idiopathic generalized epilepsy (31) | SeLECTs (24) | Other idiopathic location-related epilepsy (27)	Focal seizures (103) and generalized seizures (46)	FIAS
ADHD Diagnostic Criteria	Only inattentive, combined type	Only inattentive, only hyperactivity, combined type	Only inattentive, only hyperactivity, combined type	Only inattentive, only hyperactivity, combined type	Only inattentive, only hyperactivity, combined type	Only inattentive, only hyperactivity, combined type	Only inattentive, only hyperactivity, combined type
Age Range	8–14	6–16	9–13	5–12	5–17	6–16	7–16
Study Design	Cross-sectional	Case-control	Cross-sectional	Cross-sectional	Cross-sectional	Cross-sectional	Cross-sectional
RoB	NOS (7) NHLBI (fair)	NOS (7) ROBINS-I (serious)	NOS (8) NHLBI (good)	NOS (7) NHLBI (good)	NOS (7) NHLBI (fair)	NOS (7) NHLBI (fair)	NOS (9) NHLBI (good)
Country	Switzerland and Germany	Taiwan	Brazil	South Korea	South Korea	South Korea	USA
Sample	47 (17 Epilepsy + ADHD | 15 ADHD | 15 Controls)	122 (15 Epilepsy + ADHD | 46 Epilepsy | 61 Controls)	73 (6 Epilepsy + ADHD | 30 Epilepsy | 1 only ADHD | 36 Controls)	440 (33 Epilepsy + ADHD | 113 Epilepsy | 294 ADHD)	97 (13 Epilepsy + ADHD | 78 Epilepsy)	149 (59 ADHD + epilepsy | 90 Epilepsy)	70 (12 Epilepsy + ADHD | 21 Epilepsy | 22 ADHD | 15 Controls)
Identification	**Bechtel, 2012**	**Liu, 2011**	**Costa, 2015**	**Park, 2021**	**Lee, 2018**	**Sung-Han Kang, 2015**	**Semrud-Clikeman, 1999**
Results	No difference in IQ < 85 between ADHD patients with and without epilepsy ($p = 0.48$).	The epilepsy comorbidity + group performed worse than both comorbidity − and controls across all domains (all $p < 0.0025$). Group-by-time	ADHD + epilepsy patients had higher Stroop SEC-1 ($p = 0.012$) and SEC-3 ($p = 0.004$) scores than ADHD-only. Within ADHD + epilepsy, ADHD-I subtype	SeLECTs + ADHD patients showed longer Stroop response times (ST1 $p < 0.001$; ST2 $p < 0.001$; ST4 $p = 0.042$; total $p = 0.012$), indicating impaired attention,	Epilepsy patients, regardless of ADHD subtype, had lower IQ, verbal comprehension, reasoning, and forward span (all $p < 0.05$); no difference in	Epilepsy + ADHD patients showed lower executive and motor scores than controls and epilepsy-only ($p < 0.001$), with worse parental ratings across all	SeLECTs + ADHD patients exhibited higher CPT variability ($p = 0.008$), perseverations ($p = 0.044$), and reaction time changes ($p =$

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Table 1 (*continued*)

Results	Healthy controls outperformed unmedicated epilepsy/ADHD patients on the 2-back task ($p < 0.05$), with a trend on the 3-back ($p = 0.057$). Developmental ADHD patients had significant deficits on both tasks ($p < 0.01$ and $p < 0.05$). Medication improved 3-back task performance ($p < 0.05$).	Children with epilepsy showed more omission errors ($p = 0.020$), slower reaction times ($p = 0.032$), and higher signal detection scores ($p = 0.007$). The epilepsy + ADHD group had greater RT variability ($p < 0.05$) and reduced vigilance.	No significant differences between epilepsy patients with and without ADHD, except lower WISC Coding and Symbol Search scores in the ADHD group.	The epilepsy + ADHD group had higher auditory commission errors ($p < 0.05$) and elevated SRS scores across all subscales except Social Awareness (all $p < 0.001$).	Patients with ADHD had lower full-scale ($p = 0.014$) and verbal IQ ($p = 0.011$), more auditory omission errors ($p = 0.017$), and higher depressive symptoms ($p = 0.031$).	ADHD patients showed poorer auditory selective attention ($p < 0.020$), flanker task ($p = 0.031$), and working memory (forward $p = 0.007$; backward $p = 0.006$). In generalized seizures + ADHD, omission errors were higher ($p = 0.050$; $p = 0.003$), and working memory lower (forward $p = 0.025$; backward $p = 0.035$). SeLECTs + ADHD showed more commission ($p = 0.004$) and omission errors ($p = 0.039$).	CPS/ADHD patients performed worse than all groups on TOVA indices ($p = 0.01$); CPS and ADHD groups performed worse than controls except for commission errors ($p = 0.03$).
		interaction for executive function ($p = 0.03$) indicated worsening in comorbidity + group., driven by CPT omission and commission errors: while control and comorbidity − groups improved, the comorbidity + group showed slight worsening.	scored higher on ST-3 ($p = 0.037$) and SEC-3 ($p = 0.036$); ADHD-C + epilepsy had higher SEC-1 ($p = 0.016$), SEC-2 ($p = 0.041$), and SEC-3 ($p = 0.024$).	inhibitory control, and impulsivity.	backward span ($p = 0.09$).	executive domains ($p < 0.001$).	0.016); ADHD patients made more TMT-B errors ($p = 0.012$). After correction, WCST non-perseverative ($p = 0.031$) and TMT-A ($p = 0.033$) errors remained significant.
Cognitive Evaluation Tool	WISC-R	WISC | Wide Range Achievement Test-3 | Boston Naming Test | Expressive Vocabulary Test | Peabody Picture Vocabulary Test-III | Children's Memory Scale | Delis-Kaplan Executive Function System | CPT | Grooved Pegboard	Stroop TBAG Form | Conners Teacher Rating Scale (CTRS) | Conners Parent Rating Scale (CPRS)	WISC | SCWT	WISC-IV | Digit Span | CPT-II	WISC | Boston Naming Test | Expressive Vocabulary Test | Peabody Picture Vocabulary Test-III | Delis-Kaplan Executive Function System | Childreńs Memory Scale | CPT | Grooved Pegboard | BRIEF	Wisconsin Card Sorting Test (WCST) | WISC-IV | Matching Familiar Figures Test (MFFT) | Trail Making Test (TMT) | CPT II
Epilepsy Types	SeLECTs (2); Childhood Epilepsy with Occipital Paroxysms (CEO) (1); Symptomatic Epilepsy (1); Cryptogenic Epilepsy (6); Epilepsy with Generalized Tonic-Clonic Seizures (EGMA) (1); Childhood Absence Epilepsy (CAE) (2)	SeLECTs (2); focal (12); juvenile myoclonic epilepsy (15), idiopathic generalized epilepsy (6) and absence (7)	Not available	Self-limiting centrotemporal spike wave epilepsy (SeLECTs)	Not available	Not available	SeLECTs
ADHD Type	Only inattentive, combined type	Only inattentive, only hyperactivity, combined type	Only inattentive, combined type	Only inattentive, only hyperactivity, combined type	Only inattentive, combined type	Only inattentive, only hyperactivity, combined type	Only inattentive, only hyperactivity, combined type
Age Range	6–14	8–18	8–14	7–13	6–16	8–18	9–11
Study Design	Cross-sectional	Cohort	Cohort	Cross-sectional	Cross-sectional	Case-control	Cross-sectional
RoB	NOS (7)	NOS (9)	NOS (8)	NOS (8)	NOS (9)	NOS (7)	NOS (7)

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Table 1 (*continued*)

Results	Healthy controls outperformed unmedicated epilepsy/ADHD patients on the 2-back task ($p < 0.05$), with a trend on the 3-back ($p = 0.057$). Developmental ADHD patients had significant deficits on both tasks ($p < 0.01$ and $p < 0.05$). Medication improved 3-back task performance ($p < 0.05$).	Children with epilepsy showed more omission errors ($p = 0.020$), slower reaction times ($p = 0.032$), and higher signal detection scores ($p = 0.007$). The epilepsy + ADHD group had greater RT variability ($p < 0.05$) and reduced vigilance.	No significant differences between epilepsy patients with and without ADHD, except lower WISC Coding and Symbol Search scores in the ADHD group.	The epilepsy + ADHD group had higher auditory commission errors ($p < 0.05$) and elevated SRS scores across all subscales except Social Awareness (all $p < 0.001$).	Patients with ADHD had lower full-scale ($p = 0.014$) and verbal IQ ($p = 0.011$), more auditory omission errors ($p = 0.017$), and higher depressive symptoms ($p = 0.031$).	ADHD patients showed poorer auditory selective attention ($p < 0.020$), flanker task ($p = 0.031$), and working memory (forward $p = 0.007$; backward $p = 0.006$). In generalized seizures + ADHD, omission errors were higher ($p = 0.050$; $p = 0.003$), and working memory lower (forward $p = 0.025$; backward $p = 0.035$). SeLECTs + ADHD showed more commission ($p = 0.004$) and omission errors ($p = 0.039$).	CPS/ADHD patients performed worse than all groups on TOVA indices ($p = 0.01$); CPS and ADHD groups performed worse than controls except for commission errors ($p = 0.03$).
Country	NHLBI (good)	ROBINS-I (Moderate)	ROBINS-I (serious)	NHLBI (good)	NHLBI (good)	ROBINS-I (serious)	NHLBI (fair)
Sample	Norway 607 (593 ADHD | 14 Epilepsy + ADHD)	USA 90 (38 Controls | 24 Epilepsy | 28 Epilepsy + ADHD/AP)	Turkey 94 (47 epilepsy and ADHD | 47 ADHD)	Turkey 80 (28 epilepsy | 11 epilepsy and ADHD | 41 ADHD)	USA 91 (46 epilepsy and ADHD | 45 ADHD)	USA 137 (23 Epilepsy and ADHD | 52 Epilepsy | 62 controls)	Brazil 63 (12 epilepsy and ADHD | 11 epilepsy | 20 ADHD | 20 controls)
Identification	**Socanski et al., 2013**	**Hermann et al., 2008**	**Çam Ray et al., 2019**	**Orak et al., 2024**	**MacAllister et al., 2012**	**Hermann et al., 2007**	**Lima et al., 2018**

Caption: WISC (Weschler Intelligence Scale for Children); CPT (Conner's Continuous Performance Test); RPM (Raven's Progressive Matrices); SAT (Sustained Attention Test); TAVIS-III (Test of Visual Attention); SPT (School Performance Test); CAT (Comprehensive Attention Test); ATA (Advanced Test of Attention; BRIEF (Behaviour rating inventory of executive function); SCWT (Stroop Colour and Word Test); CBCL (Child Behavior Checklist); STAI-C (State-Trait Anxiety Inventory for Children). RoB (Risk of Bias); NOS (Nes-Castle Ottawa Scale); ROBINS-I (Risk Of Bias In Non-randomised Studies − of Interventions).

more poorly in all indices, except for reaction time.

Finally, Lima (2018) [25] used the Trail Making Test (TMT) to evaluate attention. Patients with ADHD (with or without epilepsy) had more errors on the TMT-B ($p = 0.012$). Even after adjustment for multiple comparisons, the Wisconsin Card Sorting Test (WCST-NPE) ($p = 0.031$) and TMT-A errors ($p = 0.033$) remained significant in patients with SeLECTs.

Behavioral evaluation

According to Lee (2018) [22], scores on the Children's Depression Inventory were significantly higher in the epilepsy and ADHD group compared to the epilepsy-only group ($p = 0.031$).

In the Park (2021) [21] study, the epilepsy and ADHD group and the ADHD-only group scored higher than the epilepsy-only group in all subscales of the Social Responsiveness Scale (SRS), except for Social Awareness (all $p < 0.001$).

Developmental evaluation

Only the study by Hermann (2008) [18] reported significant improvement over a 2-year interval in intelligence, academic achievement, language, memory, executive function, and psychomotor speed scores among participants, without subgroup differentiation (all p-values < 0.0001).

Cognitive outcomes based on Wechsler scales

Table 2
Summary of findings for Wechsler-based cognitive outcomes.

Outcome	N° of studies	Findings summarized	Risk of bias	Quality of the evidence (GRADE)	Comment
Complex Attention	4	Mild to moderate impairment in two studies, others preserved	1 high risk; 1 moderate risk; 2 low risk	Very Low	Due to inconsistency and imprecision
Executive Function	8	Mild impairment in five studies, others preserved	1 high risk; 3 moderate risk; 4 low risk	Very Low	Due to imprecision
Perceptual-Motor Function	8	Mild to moderate impairment in four studies, others preserved	1 high risk; 3 moderate risk; 4 low risk	Very Low	Due to imprecision
Language	8	Mild to moderate impairment in three studies, others preserved	1 high risk; 3 moderate risk; 4 low risk	Very Low	Due to imprecision
Social cognition	8	Mild to moderate impairment in three studies, others preserved	1 high risk; 3 moderate risk; 4 low risk	Very Low	Due to imprecision
FSIQ	9	Mild impairment in three studies, others preserved	2 high risk; 3 moderate risk; 4 low risk	Very Low	Due to risk of bias and imprecision

Caption: Summary of findings derived from the nine studies that assessed DSM-5 cognitive domains using Wechsler-based instruments (WISC, WAIS-IV, or WASI) in the epilepsy + ADHD group. Certainty of evidence was evaluated using the GRADE approach.

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Table 3
Pharmacological treatment and medication control across included studies.

Study	Antiepileptic medications (AEDs)	AEDs use status	ADHD medications	ADHD medication use status	Control for medication effects
Bechtel, 2012	AEDs with no substantial cognitive impact (e.g., oxcarbazepine, valproate)	All patients under stable AED treatment, seizure-free ≥ 6 months	Methylphenidate (MPH)	53 % of Epilepsy + ADHD and 93 % of Developmental ADHD patients were habitual users; all tested with and without MPH	Cross-over design; 24 h washout before "off-medication" session; ≥ 14 days between sessions
Liu, 2011	39 carbamazepine; 14 valproate; 1 phenobarbital	58 using AEDs	MPH	Three participants with epilepsy and ADHD used MPH currently	NM
Costa, 2015	NM	86.1 % used one AED and 11.1 % used more than one.	NM	NM	NM
Park, 2021	Exclusion criteria	NM	Exclusion criteria	NM	NM
Lee, 2018	Exclusion criteria	NM	NM	NM	NM
Sung-Han Kang, 2015	NM	94 used single AED and 19 used dual AED	NM	NM	NM
Semrud-Clikeman, 1999	Carbamazepine (CBZ)	78 % were using one AED	MPH	All participants were administered MPH in the second appointment. Before that, MPH-naive participants.	Exclusion criteria
Socanski et al., 2013	Valproate (VPA), carbamazepine (CBZ), oxcarbazepine (OXC), lamotrigine (LTG), levetiracetam (LEV), sulthiame (SUL), topiramate (TPM), ethosuximide (ESM)	10 used AED monotherapy and 2 used AED polytherapy	MPH	All participants with epilepsy + ADHD used MPH	NM
Hermann et al., 2008	NM	39 single AED and 1 dual AED	NM	NM	NM
Çam Ray et al., 2019	Valproate; Carbamazepine; Other AED	80.9 % used valproate; 12.8 % used carbamazepine and 5 other AED	NM	NM	NM
Orak et al., 2024	Levetiracetam; Cabamazepine; Others not specified	All epilepsy + ADHD used single AEDs. 77 % used single AEDs and 23 % used dual AEDs	NM	71.4 % used psychostimulant	NM
MacAllister et al., 2012	NM	Those with ADHD-I, 48.4 % single AED. With ADHD-C, 60 % used single AED.	NM	NM	NM
Hermann et al., 2007	NM	60 patients used at least one AED and 3 at least two AEDs.	MPH, Atomoxetine, Amphetamine	In epilepsy + ADHD group, 10 of 23 participants were using ADHD medication	NM
Lima et al., 2018	NM	NM	MPH users excluded from the study	NM	Inclusion criteria included not using psychiatric drugs

Caption: Abbreviation: NM = Not mentioned.

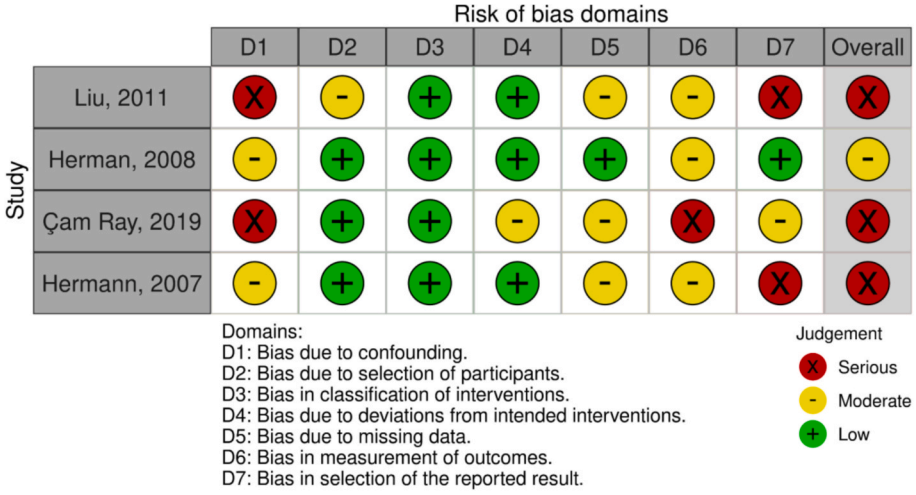


Fig. 2. Risk of bias assessment using the ROBINS-I tool. Legend - Traffic-light plot summarizing the risk of bias assessment for the included non-randomized studies, as evaluated with the ROBINS-I tool. Each domain is color-coded as low risk (green), moderate risk (yellow) or serious risk (red). Inter-rater reliability between reviewers was moderate (κ = 0.50), and final judgments were reached by consensus.

In accordance with the six cognitive domains defined by the DSM-5 (complex attention, executive function, learning and memory, language, perceptual–motor function, and social cognition) we analyzed the performance patterns derived from the Wechsler-based instruments (WISC, WAIS-IV, and WASI), as described in the Methods section.

A comprehensive heat map (Fig. 3) was developed to integrate and

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Table 4
Mapping of Wechsler subtests and composite indices to DSM-5 cognitive domains.

Cognitive Assessment Tool	Index / Subtest	Corresponding DSM-5 Cognitive Domain(s)
WISC	Verbal Comprehension	Language; Social Cognition
	Perceptual Organization / Fluid Reasoning	Perceptual-Motor Function; Executive Function
	Working Memory / Attention	Complex Attention; Executive Function
	Processing Speed	Complex Attention; Perceptual-Motor Function
WASI	Vocabulary	Language; Learning and Memory
	Similarities	Executive Function; Language
	Block Design	Perceptual-Motor Function; Executive Function
	Matrix Reasoning	Executive Function; Perceptual Motor Function
WAIS-IV	Verbal Comprehension Index (VCI)	Language; Social Cognition
	Perceptual Reasoning Index (PRI)	Perceptual-Motor Function; Executive Function
	Working Memory Index (WMI)	Complex Attention; Executive Function
	Processing Speed Index (PSI)	Complex Attention; Perceptual-Motor Function

Caption: Correspondence between Wechsler-based measures (WISC, WAIS-IV, and WASI) and the six cognitive domains defined by the DSM-5: complex attention, executive function, learning and memory, language, perceptual–motor function, and social cognition. This mapping guided the domain-level synthesis and the construction of the cognitive heat map.

visually represent the main cognitive findings across the included studies, illustrating the severity and distribution of impairments across these domains for the epilepsy + ADHD, epilepsy only, and ADHD only groups. Each cell in the heat map corresponds to the level of alteration in a specific cognitive domain, categorized according to standardized Wechsler score interpretation criteria. [16].

In the epilepsy + ADHD group, the analyses indicated a heterogeneous but notable pattern of deficits. In the domain of complex attention, one study (Costa, 2015) [24] reported moderate impairment, one reported mild impairment, two found preserved performance, and five studies observed performance within or above the average range. Executive function was the most consistently affected domain, with five studies identifying mild impairment, while three reported preserved function and one did not assess this domain. Learning and memory were not evaluated by the included Wechsler-based tests. Regarding language, one study (Orak, 2024) [26] identified moderate impairment, two reported mild alterations, five indicated preserved function, and one did not evaluate this domain. For perceptual–motor function, one study (Costa, 2015) [24] reported moderate alteration, three observed mild impairment, four found preserved performance, and one did not assess it. In the domain of social cognition, one study (Orak, 2024) [26] reported moderate alteration, two studies mild alteration, five preserved function, and one did not assess this construct. Full-Scale IQ (FSIQ) was mildly reduced in three studies and preserved in the remaining six.

In the epilepsy-only group, the cognitive profile appeared more circumscribed, with impairments generally mild and less widespread. Complex attention was mildly altered in one study (Costa, 2015) [24], preserved in two, and not assessed in five. Executive function demonstrated mild impairment in two studies (Costa, 2015; Semrud-Clikeman, 1999) [17,24], while five reported preserved performance and one did not assess this domain. Learning and memory were not evaluated. For language, one study (Orak, 2024) [26] showed mild alteration, six reported preserved performance, and one did not assess it. Perceptual–motor function exhibited mild alteration in two studies (Costa, 2015; Semrud-Clikeman, 1999) [17,24], preserved in five, and not assessed in one. In social cognition, mild alteration was reported by one study (Orak, 2024)[26], with preservation in six and non-assessment in one. FSIQ was mildly reduced in Semrud-Clikeman (1999) [17], whereas all other studies indicated preserved overall intellectual functioning.

Finally, for the ADHD-only group, all assessed domains demonstrated preserved performance across the studies, with no evidence of cognitive impairment in the parameters evaluated by the Wechsler instruments.

Taken together, these findings illustrate that the co-occurrence of epilepsy and ADHD is associated with broader and more frequent cognitive alterations, particularly in executive and attentional domains, while isolated epilepsy presents a milder, domain-specific profile, and

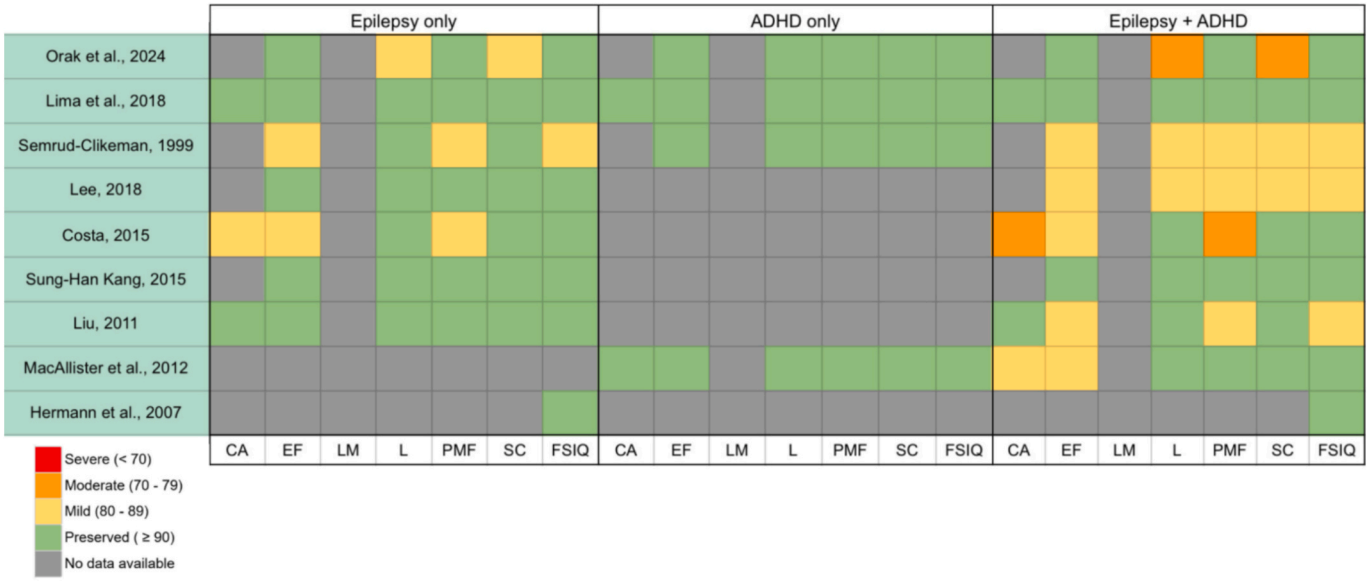


Fig. 3. Heat map representation of cognitive domains across clinical groups. Legend: Heat maps illustrating the degree of impairment across cognitive domains in three groups: comorbid epilepsy + ADHD, epilepsy only, and ADHD only. Domains were classified according to performance level (preserved, mild, moderate, or severe impairment) based on Wechsler-derived measures. Abbreviations: CA, Complex Attention; EF, Executive Function; LM, Learning and Memory; L, Language; PMF, Perceptual-Motor Function; SC, Social Cognition; FSIQ, Full Scale IQ.

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ADHD alone shows preserved global cognitive performance within the evaluated parameters.

In the epilepsy-only group, the study by Lee et al. (2018) [22] reported significant differences in cognitive performance according to epilepsy subtype. Participants with Idiopathic Generalized Epilepsy (IGE) demonstrated lower scores in overall intellectual functioning (FSIQ = 89.0 ± 17.6) and nonverbal reasoning (PIQ = 88.9 ± 16.3) compared with those with Idiopathic Localization-Related Epilepsy (ILRE) (FSIQ = 96.3 ± 14.8; PIQ = 97.0 ± 16.4). These findings suggest that, even in the absence of ADHD comorbidity, the type of epilepsy itself may influence global cognitive performance. However, for the purpose of standardization in the heat map synthesis, the epilepsy-only group from this study was represented as having preserved domains, as the observed differences were restricted to a single epilepsy subtype, whereas the remaining subtypes assessed in the same study showed preserved values, and the degree of alteration was borderline. This clarification is provided here to ensure transparency and methodological consistency in the interpretation of results.

Taken together, this Wechsler-based synthesis provides a standardized and integrated overview of the cognitive profiles across clinical groups, facilitating the visualization of domain-specific vulnerabilities and highlighting the consistent involvement of attentional and executive processes in pediatric epilepsy with comorbid ADHD.

3.4. Methodological notes and analytical limitations

It is important to acknowledge several methodological limitations that may influence the interpretation of the synthesized cognitive findings. First, not all studies administered the Wechsler-based instruments to their entire participant samples, resulting in partial data availability across groups. Second, because each Wechsler scale comprises multiple subtests that capture distinct nuances within the same cognitive domain, occasional discrepancies were observed between subtest outcomes within a single domain. To ensure standardization in the heat map representation, any domain presenting at least one altered subtest was classified as impaired, and the level of alteration (mild, moderate, or severe) was determined by the most pronounced deficit observed. Third, only nine of the fourteen studies included in the systematic review were represented in the heat map, as only eleven employed the WISC, WAIS-IV, or WASI. Of these, two could not be standardized: the study by Socanski et al. did not provide numerical results for group comparisons, and Hermann et al. (2008) [18] reported only raw scores, precluding conversion to standardized values. Fourth, data for the epilepsy-only comparator group were available in eight studies, whereas only four studies provided data for the ADHD-only comparator group. These methodological constraints should be taken into account when interpreting the relative distribution and severity of cognitive impairments across the clinical groups represented in the heat map.

4. Discussion

This study aimed to evaluate cognitive performance in pediatric patients with epilepsy and comorbid ADHD. Our systematic review indicated that children with both conditions exhibited more severe impairments than those with epilepsy alone, ADHD alone, or healthy controls. These findings highlight that ADHD comorbidity in pediatric epilepsy is associated with greater neurocognitive deficits.

Although the pathophysiological basis for the comorbidity between these two conditions remains unclear, their correlation is well established. Some studies estimate a global prevalence of epilepsy in pediatric patients ranging from 0.5 % to 1 % [30]. For ADHD, an estimated prevalence of 8.0 % is reported among children and adolescents [5]. Among patients with epilepsy, the prevalence of ADHD may range from approximately 30.7 % to 49.2 % [6,23].

A degree of consistency can be observed in the results across studies included in this review. Nearly all articles found a significant cognitive impact in patients with both ADHD and epilepsy, although the severity and specific cognitive domains affected varied. For instance, Park et al. [21] demonstrated negative effects on inhibitory control, aggressive behavior, and parental stress. Orak et al. [26] reported greater impairments in verbal intelligence and sustained attention. Kang et al. [23] found alterations in selective attention, flanker test results, spatial working memory, sustained attention, and interference, depending on the subtype of epilepsy (generalized vs. focal). These findings underscore the importance of analyzing cognitive outcomes in relation to the epilepsy subtype when ADHD is present.

Neuroimaging studies support these findings by identifying not only brain alterations associated with epilepsy alone, but also significantly intensified changes in the presence of ADHD. For example, Kim et al. [31] reported anatomical abnormalities in patients with SeLECTs, particularly in regions associated with executive function. Lima et al. [25] further showed that patients with both SeLECTs and ADHD exhibited increased cortical thickness in the left caudal anterior and posterior cingulate gyri, and greater volume in the left pars opercularis gyrus, compared to those without ADHD. These regions, which are functionally connected to the frontal lobe, are involved in executive functions such as working memory, decision-making, the limbic system, selective attention, inhibitory control, and verbal fluency. Collectively, these neuroimaging findings reinforce the notion that ADHD comorbidity exacerbates cognitive deficits in pediatric epilepsy.

This review presents several limitations. First, despite an extensive search strategy, only a small number of articles met the inclusion criteria. Randomized clinical trials were not included, primarily due to the lack of eligible studies of this type. Most included studies had small sample sizes, which limited the evaluation of cognitive performance related to ADHD comorbidity across different epilepsy subtypes and may have impacted the robustness of the results. Also, although we did not perform a quantitative *meta*-analysis due to the heterogeneity of study designs, populations, and outcome measures across the included studies, the possibility of reporting bias cannot be excluded. We acknowledge that small-study effects and selective reporting may have influenced the available evidence, and these factors should be considered when interpreting the findings of this review.

Nonetheless, a notable consistency in findings was observed. Most studies demonstrated a significant relationship between cognitive decline and the coexistence of ADHD and epilepsy. Two studies, however, did not report a statistically significant association between cognitive performance and the presence of both conditions. Socanski et al. and Costa et al. [15,24], both cross-sectional in design, found no significant differences in IQ among ADHD patients with and without epilepsy. Similarly, Bechtel et al. [29] reported that both the ADHD-only and ADHD + epilepsy groups performed comparably worse than controls on high cognitive load tasks. A notable finding in this study was the significant improvement observed in both groups with the use of methylphenidate, suggesting that early pharmacological intervention may positively influence cognitive development.

Given the substantial cognitive impact of this comorbidity, implementing systematic screening for ADHD in all children with epilepsy becomes a necessary intervention. Based on expert consensus, screening should be performed for all children with epilepsy starting at age 6, or at the time of diagnosis if older, and repeated annually. The process can be initiated using validated ADHD rating scales administered to parents and teachers. Among the available tools, the Strengths and Difficulties Questionnaire (SDQ) is supported by Level B evidence for its ability to predict an ADHD diagnosis in this population. Other instruments, such as the Swanson, Nolan, and Pelham-IV (SNAP-IV) questionnaire, have also been used. A positive result on one of these screening tools should then lead to a comprehensive diagnostic evaluation by a qualified healthcare professional with expertise in ADHD, ensuring that children receive a timely and accurate diagnosis, which is the first step toward targeted therapeutic interventions. [32] Also, some authors have

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suggested a bidirectional relationship between ADHD and epilepsy, in which attentional and executive dysfunction may both precede and follow the onset of seizures. In this context, clinical vigilance is warranted in children diagnosed with ADHD, particularly those with the inattentive subtype who present with paroxysmal events that could indicate seizure activity or who exhibit a poor or atypical response to stimulant medications [33]. This ADHD clinical profile suggests that further neurological evaluation and screening for epilepsy may be appropriate.

In conclusion, this review supports a strong correlation between ADHD and epilepsy and suggests that the coexistence of both conditions is directly associated with poorer cognitive outcomes in pediatric patients when compared to those with only one of the conditions or healthy controls. Early identification is essential, as it enables timely intervention to support cognitive development and minimize long-term impacts on affected children. This finding raises important questions regarding the pathophysiological mechanisms underlying these cognitive impairments, whether shared or distinct, and highlights the urgent need for further research to clarify this relationship, develop more effective treatments, and improve long-term outcomes for these patients.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used *ChatGPT (OpenAI, GPT-5)* and *Gemini (Google DeepMind)* to assist in the organization, synthesis, and language refinement of the manuscript. These tools were employed to support tasks such as improving clarity, coherence, and academic style, and to help standardize methodological descriptions and summaries of results based on data extracted and verified by the authors. After using these tools, the authors carefully reviewed, edited, and validated all content to ensure accuracy, originality, and adherence to the journal's ethical and scientific standards. The authors take full responsibility for the integrity and validity of the published article.

CRediT authorship contribution statement

Laura Zaffari Leal: Writing – original draft, Formal analysis, Data curation. **Inácio Kehl:** Writing – review & editing, Formal analysis. **Natália Donati Polesello:** Writing – review & editing, Formal analysis. **Yolanda Aquino de Souza:** Writing – review & editing, Formal analysis. **Catarina Hauser Schmitz:** Writing – review & editing, Formal analysis. **Thiago Wendt Viola:** Writing – review & editing, Data curation. **Magda Lahorgue Nunes:** Writing – review & editing, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.yebeh.2025.110852.

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